Sukhoi Su-57 (Felon): A Fifth-Generation Multirole Stealth Fighter - V1.0

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P.S Also, I am 17 and my parents will kick me out if I get in trouble with the government, please.

Abstract

The Sukhoi-57 (Su-57) represents Russia's entry into fifth-generation fighter aircraft, combining stealth, advanced electronic warfare capabilities, super-maneuverability, and advanced avionics.This paper presents an aerodynamic analysis of the Su-57, with particular focus on propulsion, stability derivatives, manuverability and avionic architecture.

1 Acknowledgements

The author thanks the DCS World Su-57 development community for providing the impetus for this research, and acknowledges NASA DATCOM (NASA Technical Report 19950005762) as the primary methodological reference for stability derivative derivations.

2 Introduction

The Sukhoi-57, named "Felon" by NATO is Russia's first operational fifth-generation, single-seat, twin-engine(with thrust vectoring) multirole fighter designed for air superiority, ground strikes, and stealth operations

3 Development Background

The Su-57 was developed under the PAK FA program with the original designation "T-50", initiated in the late 1990s. Its first flight took place in 2010, and it entered limited service around 2020. Different variants of the Felon includes, the production variant being the Su-57(with the AL-41F1 (117S) engines), the newer and production variants include the Su-57E, and the Su-57M (with improved Izdeliye 30 engines and upgraded avionics)

4 Aerodynamic Design

The aircraft features a blended wing-body configuration (reducing drag and radar cross section). The inclusion of thrust vectoring nozzles enables extreme maneuverability (Pugachev's Cobra, Kulbit), including high angle-of-attack stability. The Felon also uses the LEVCON vortex controllers to

manipulate wingtip vortices to generate (or increase) lift and increase AOA (Angle Of Attack).

5 Stealth Features

Stealth is achieved through shaping, internal weapons bays, and radar-absorbing materials. While not as stealth-focused as Western counterparts, it balances stealth with maneuverability.

6 Propulsion and Performance

The Su-57 is powered by AL-41F1 engines, with future upgrades to Izdeliye 30 engines. It is capable of supercruise and reaches speeds of approximately Mach 2.

7 Aerodynamic Modeling

7.1 Calculation Assumptions Based On Public Data:

geometry derived estimates:

- Length: 20.1m
- Wingspan: 14.1m
- Height: 4.6m
- Wing Area: 78.8m^2 ‘1
- Empty Weight: 18,500kg
- Gross Weight: 26,700kg (Normal Config), 29,770kg (Full Load)
- Fuel Capacity: 10,300kg
- Powerplants: 2x AL-41F1 - 88.3kN Dry, 142.2kN Afterburner, 147.1kN Max Emergency Power
- Max Speed: Mach 2 at Altitude, Mach 1.1 at Sea Level

- Supercruise Speed: Mach 1.3 at High Altitude
- Wingspan: 14.1m
- Sweep Angle: 48° LE, 14° TE
- Aspect Ratio: 2.49
- Oswald Efficiency Factor: 0.9
- Zero-Lift Drag Coefficient: 0.015-0.025
- Drag Coefficient: 0.0005 (Clean Aircraft) — 0.015 (Bay Doors Open)
- Max Roll Rate: 220°/s
- Instantaneous Turn Rate: 28-39°/s
- Corner Velocity: 650-830km/h (350-450kts)
- Max AOA before Departure: 70-90°
- C_{l_β} : -0.15 to -0.25/rad (Dihedral Effect)
- C_{n_β} : 0.08 to 0.15/rad (Directional Stability)
- $C_{l_{\delta a}}$: -0.4 to -0.6/rad (Aileron/Flaperon Roll Power)
- $C_{n_{\delta r}}$: 0.2 to 0.35/rad (Rudder Yaw Power)
- C_{l_p}, C_{n_r}: -0.5 to -0.8/rad (Roll/Yaw Damping)

7.2 Stability Derivatives Comparison

The following table presents a comparison between values sourced from open literature and those derived independently using NASA DATCOM methodology applied to publicly available Su-57 geometry.

NASA METHODOLOGY LINK:
`https://ntrs.nasa.gov/api/citations/19950005762/downloads/19950005762.pdf`

Table 1: Stability Derivatives: Open Literature vs. DATCOM Derived

Derivative	**Open Literature**	**DATCOM Derived**	**Physical Meaning**
C_{L_α}	nil	2.728/rad	Lift curve slope
C_{m_α}	-0.02 to 0.05/rad	-0.055/rad	Longitudinal instability
C_{m_q}	-0.8 to -1.2/rad	-1.346/rad	Pitch damping
C_{l_β}	-0.15 to -0.25/rad	-0.304/rad	Dihedral effect
C_{n_β}	0.08 to 0.15/rad	0.242/rad	Directional stability

Note: Discrepancies between open literature and DATCOM derived values are attributable to geometric estimation uncertainty and excluded LEVCON contributions. Control surface derivatives ($C_{l_{\delta a}}$, $C_{n_{\delta r}}$, C_{l_p}, C_{n_r}) are retained from open literature as first-principles derivation requires detailed control surface geometry unavailable in public domain.

Weapon Bay Shifts:

- $\Delta C_{x,\text{bay}}$: 0.015 to 0.035
- $\Delta C_{m,\text{bay}}$: 0 to 0.015
- $\Delta C_{y,\text{bay}}$: < 0.005
- ΔC_{y_α} / lift change: -0.005 to -0.01

7.2.1 LIFT

The standard Lift(L) equation:

$$L = \frac{1}{2}\rho V^2 S C_L$$

- where L is the lift Force, ρ is the air density(Here we are going to take it a 1.225 Kg/m^3 sea level), V is the velocity of the aircraft, S is the Wing area (Basically the whole plane), and C_L is the lift coefficient which depends on the AOA and wing design.
- Now this gets even more interesting for the Su-57 and most 5th gen aircraft, From the above equation, we know that Lift depends on airflow, but for aircraft like the Su-57, it seems that it stays airborne and controlled even during Cobra maneuvers, cause clearly, for the above

equation to work, it needs smooth airflow (attached flow) and lift is generated by the pressure difference between the bottom of the wing (high pressure) and the top of the wing (low pressure). The question arises "How and why does this happen?" Here's why:

- **During the Cobra:**
  - **High AOA:**

    Angle of attack becomes very high
    Airflow separates from the wing
    Standard lift equation starts to break down

  - **Vortex Lift:**
    Vortices form at the edge of both the wings, the core of these vortices have a very low pressure zone, above the wing, along leading edges and around the fuselage-wing junction.

    Rotating airflow at the edges of the wings, that has low pressure at it's core, this vortex "sticks" over the wing and creates localised suction, which produces lift.

    The Su-57 is especially good at vortex control because of it's LEVCONs (Leading Edge Vortex Controllers).

  - **Thrust Vectoring:**
  - The Su-57 has 3D thrust vectoring, that can enable it to redirect thrust for control, thrust becomes a major substitute for lift+control.
  - **High C_L region:**
  - Before full stall, the aircraft can increase AOA
  - This increases C_L, because of high area of aircraft(flat, 'leaf' like profile) calculated for Lift coefficient, the Su-57 can operate at very high C_L.

– **Lift Coefficient C_L generated by the Su-57:**

$$L = \frac{1}{2}\rho V^2 S C_L$$

– In level flight

$$L = W$$
$$C_L = \frac{2W}{\rho V^2 S}$$

$W = 261,927N$
$v = 205m/s$
$\rho = 1.255Kg/m^3$
$S = 78.8m^2$

$$C_L = 0.26$$

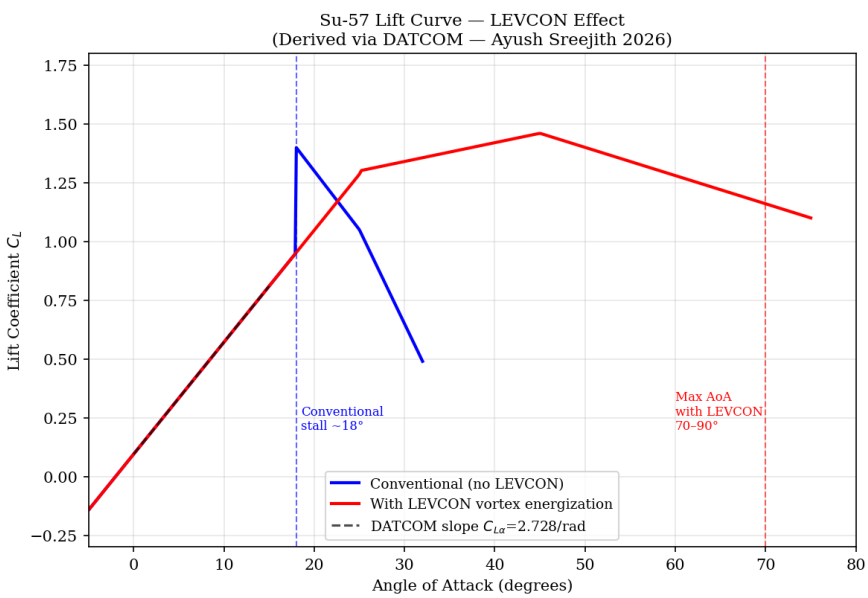

Figure 1: Su-57 Lift Curve — LEVCON Extension Effect

7.2.2 DRAG POLAR

The standard Drag Polar(C_D) equation:

$$C_D = C_{DO} + KC_L^2$$

- Where C_D is the Total drag coefficient, C_{DO} is the parasite drag coefficient (drag at zero lift, includes skin friction and form drag), C_L is the Lift coefficient of the aircraft.
- The induced drag factor (K) is given by:

$$K = \frac{1}{\pi e AR}$$

- Where e is the Oswald efficiency factor (we take it as 0.9 for the Su-57), AR is the Aspect Ratio of the wing($AR = b^2/S$)

Since there is always an exception case in physics, C_D At high speeds:

$$C_D = C_{DO} + KC_L^2 + C_{D_{wave}}$$

- $C_{D_{wave}}$ accounts for the sharp increase in drag of an aircraft at supersonic speeds, it is induced by the shockwaves from the "sonic boom" produced by the aircraft, $C_{D_{wave}}$ depends upon Mach number M, when M is greater than M_{crit}
- **Drag polar below supersonic:**
  $C_{DO} = 0.02$
  $C_L = 0.26$
  $e = 0.9$(const.)
  $AR = 2.49$(const.)

$$C_D = 0.0297$$

7.3 Weapon Bay Aerodynamic Effects

Opening the internal weapons bays introduces measurable aerodynamic penalties. The primary effects are:

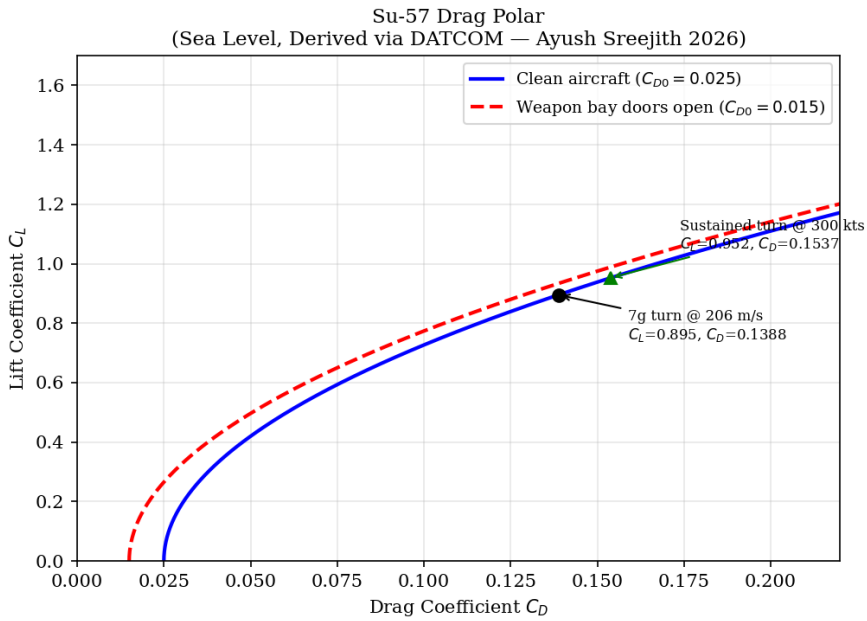

Figure 2: Su-57 Drag Polar — Clean vs Weapon Bay Doors Open

7.3.1 Drag Increase

When weapon bay doors open, the zero-lift drag coefficient C_{D0} increases from 0.025 (clean) to approximately 0.015–0.035, due to:

- Cavity drag from open bay volume
- Disrupted boundary layer at bay opening
- Increased form drag from door geometry

From the weapon bay shift data:

- $\Delta C_{x,\text{bay}}$: 0.015–0.035 (axial force coefficient increase)
- $\Delta C_{m,\text{bay}}$: 0–0.015 (small pitch moment shift)
- $\Delta C_{y,\text{bay}}$: < 0.005 (negligible lateral force)

Taking $\Delta C_{x,\text{bay}} = 0.025$ (midpoint), total drag during bay open:

$$D_{\text{bay}} = (C_{D0} + \Delta C_{x,\text{bay}} + C_{Di}) \times q \times S$$

$$= (0.025 + 0.025 + 0.1137) \times 25,992 \times 78.8$$

$$= 0.1637 \times 25,992 \times 78.8 \approx \mathbf{335\ kN}$$

Compared to clean configuration drag of 60.6 kN — bay doors open increases drag by approximately **453%** at combat speed.

7.3.2 Operational Implication

This drag penalty explains why the Su-57's weapon release sequence is designed for minimum door-open time. The side quick-launch bays demonstrated at Dubai Airshow 2025 use a rapid extend-retract mechanism specifically to minimise this drag exposure window during within-visual-range engagements. The main tandem bays are used primarily in BVR engagements where the aircraft can afford brief speed reduction during launch.

The drag polar comparison between clean and bay-open configurations is shown in Figure 2.

- **Drag polar at supersonic speeds:**
- And how exactly does Drag Polar scale at supersonic speeds?
- K becomes:
$$K = \frac{1}{\pi e AR \beta}$$
- where β:
$$\beta = \sqrt{M^2 - 1}$$
- so, from the above equations, at supersonic speeds, β increases, K decreases, implies that lift becomes more "cheaper", implying that less surface area of the plane is needed for lift.
- At these speeds, the main goal is to engineer these aircraft to reduce wave drag, that is why many 5th gen fighters have sharp edges on it's airframe and in this case, thats why the Su 57 has swept wings(48° LE)

- Drag coefficient at supersonic speeds (assumptions at supercruise at Mach 1.3):

  $C_{DO} = 0.02$
  $C_{wave} = 0.02$
  $C_L = 0.015$
  $K = 0.17$
  $\rho = 0.38kg/m^3$
  $V = 440m/s$
  $S = 78.8m^2$
  $T = 176KN$ (dry, at supercruise)

$$C_D = 0.02 + 0.025 + (0.17 \times 0.015^2)$$

$$C_D \approx 0.049$$

- Drag:

$$D = \frac{1}{2}\rho V^2 C_d A$$

$$D = 0.5 \times 0.38 \times 440^2 \times 78.8 \times 0.049$$

$$D = 142,000N$$

- Excess Thrust:

$$T_{Excess} = T - D$$
$$= 176 - 142$$
$$T_{Excess} \approx 30KN$$

- Aircraft is stable with no huge acceleration, sustainable supercruise with enough margin

7.3.3 DRAG FORCE

The standard Drag force (D or F_d) equation:

$$D = \frac{1}{2}\rho V^2 C_d A$$

- **Values (subsonic):**
  $\rho = 1.225$ (At sea level)
  $V = 206$ m/s
  $S = 78.8m^2$
  $C_D = 0.0297$ (From drag polar)

- Dynamic pressure (q):

$$q = \frac{1}{2}\rho V^2$$
$$q = 25900$$

$$D = qSC_d$$
$$D \approx 60,600$$

- At sea level, traveling at subsonic speed of 206 m/s, Drag force of the Su-57 is approximately 60.6 KN.

7.3.4 Turn Rate:

Instantanious turn rate equation:

$$\omega = \frac{g\sqrt{n^2 - 1}}{V}$$

- where ω is the Inst. turn rate, g is the gravitational force of the earth (Here we are going to take it a 9.81 m/s), n is the load factor (G's) of the aircraft, V is the True Airspeed (TAS or corner velocity).

- **At Standard Structural Limit:**
  $n = 9$ (standard structural limit)
  $g = 9.81$ m/s
  $V = 206$ m/s (general combat speeds)

$$\omega = \frac{9.81 \times \sqrt{9^2 - 1}}{206}$$
$$\omega = 0.425 \text{ rad/s}$$
$$\omega = 24.4^\circ/s$$

At 7G's (Realistic):

$$\omega = \frac{9.81 \times \sqrt{7^2 - 1}}{206}$$

$$\omega = 0.329 \text{ rad/s}$$

$$\omega = 18.9^\circ/s$$

- This is baseline turn rate and assumes no Thrust Vectoring and LEVCON, with the LEVCON and TVC config., the pure structural limit of the Su-57 is much higher,upto $\omega = 30.4^\circ/s$ keeping velocity const,(inst. turn rate scales with decrease in velocity), with TVC and LEVCON's (AB), and upto $n = 11.2$ (11 G's), but at that level of G's, the pilot turns into human mush, that is why TVC and LEVCON consideration is not of big significance when calculating inst. turn rate.

Sustained Turn rate:

- Here the LEVCON and TVC system of the Su-57 comes into play,
- **Assumed values:**
  $V = 154.3$ m/s
  $q = 14,583$ Pa (Calculated from dynamic pressure equation at sea level)
  $T_{total} = 2 \times 88,300 = 176,600$ (engine thrust)
  $C_D = T/qS = 0.1537$
  $C_{Di} = 0.137 - 0.025 = 0.1287$ (Parasitic drag=0.025)
  $C_L = \sqrt{0.1287 \times \pi \times 2.49 \times 0.9} = 0.952$

  $$n_{base} = \frac{C_L qS}{mg}$$

  $$n_{base} = \frac{0.952 \times 14,583 \times 78.8}{261,927}$$

  $$n_{base} \approx 4.18 \text{ g}$$

  This means that the g load that the engine alone can support is 4.18 g.
- Including TVC and LEVCON's:
  $\delta n_{LEVCON} = +0.627$ g (computed taking LEVCON=+15% C_L)
  $\delta n_{TVC} = +0.174$ g (computed taking TVC=+15%, dry)

  $$n_{total} = 4.18 + 0.627 + 0.174 = 4.98g$$

$$\omega = \frac{9.81 \times \sqrt{4.98^2 - 1}}{154.3}$$
$$\omega = 17.8°/s$$

- Published figures of $25 - 35°/s$ cannot be achieved at optimal rate speeds of 300 knots, even with TVC and LEVCON augmentation. Achieving 25° would require about 427 KN of thrust which is about 50% more than the published AL-41F1 figures. This discrepancy may be due to the conservative reporting of propulsion data, or geometric estimation uncertainty or both.

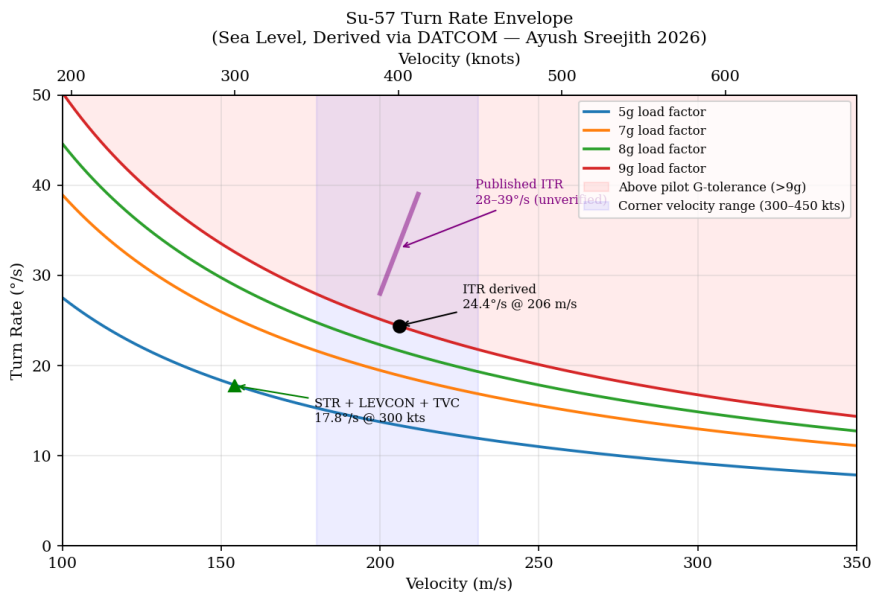

Figure 3: Su-57 Turn Rate Envelope — Derived vs Published

7.3.5 ROC(Rate Of Climb)

Standard Rate Of Climb equation:

$$ROC = \frac{(T - D) \times V}{W}$$

- Where T is the thrust of the aircraft, D is the drag force, W is the weight of the aircraft.V is the velocity of the aircraft.(we take standard

velocity as 206 m/s)

- To calculate Thrust:

$$TWR = \frac{T}{W}$$

$$TWR \times W = T$$

$$T = 1.09 \times 261,927 \text{ (taking Normal gross weight as 26,700 Kg)}$$

$$T = 285,500 \text{ N (Total thrust)}$$

D=60,800 N
V=206 m/s
W=261,927 N

$$ROC = \frac{(285,500 - 60,800) \times 206}{261,927}$$

$$ROC \approx 176.7\text{m/s}$$

- The Su-57's Rate of Climb was derived using sea level air density, with standard velocity of 206 m/s, the derived ROC is approximately 176.7 m/s (34,780 ft/min), even with max emergency power,ROC goes up to about 183.6 m/s , which is exceptional for an aircraft but widely circulated figures of 350-384 m/s (which would need a 274 kN engine) is not accurate based on this calculation. This is consistent with the pattern observed in turn rate analysis, integrated performance figures in open literature significantly exceed values derivable from published component specifications.

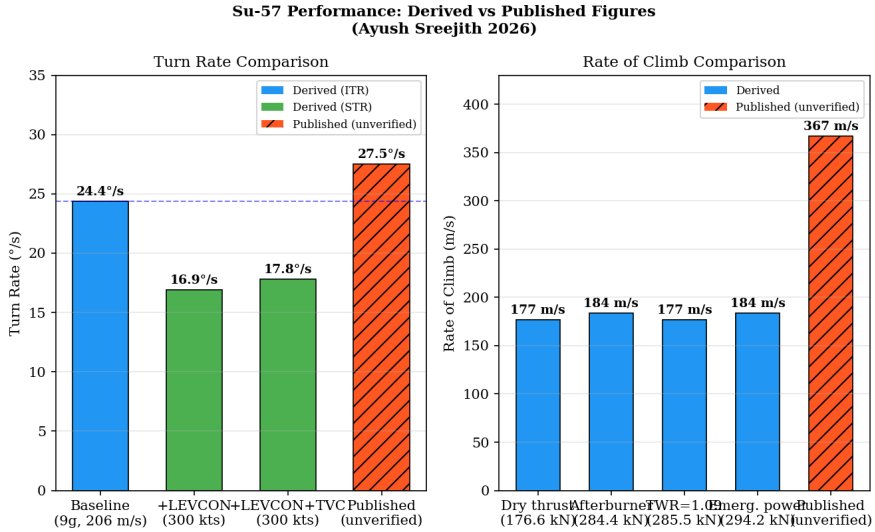

Figure 4: Performance Discrepancy: Derived vs Published Figures

8 Avionics and Systems

The Su-57 incorporates the **Sh121 integrated radar system**, comprising three primary components:

- **N036 Byelka AESA Radar** — Forward-facing X-band active electronically scanned array with approximately 1,526 transmit/receive modules, providing long-range target detection, tracking, and engagement capability.
- **N036B Side-Facing Arrays** — Two lateral AESA arrays providing wide-area situational awareness and expanding the overall radar field of regard beyond the forward sector.
- **N036L L-Band Wing Arrays** — Leading edge L-band arrays providing counter-stealth detection capability against low-observable aircraft, exploiting resonance effects at longer wavelengths.

The radar system operates in **Low Probability of Intercept (LPI)** modes, reducing detectability by enemy radar warning receivers. It supports simultaneous multi-target tracking and engagement, and is integrated with

the **L402 Himalayas** electronic warfare suite for passive detection, jamming, and situational awareness.

The cockpit features advanced automation reducing pilot workload through sensor fusion — combining radar, infrared search and track (IRST), and electronic warfare data into a unified tactical picture.

8.1 Radar Cross Section (RCS)

The Su-57 incorporates radar-absorbent materials (RAM), internal weapons bays, and edge-aligned surfaces to reduce its radar signature. It is important to note that RCS is not a single fixed value — it varies with aspect angle, frequency band, sensor polarization, weapons carriage state, and RAM condition.

8.1.1 Published RCS Values (X-band, Head-on, Clean Configuration)

Based on Russian state sources and program documentation, the Su-57 RCS goal was stated as 0.1–1.0 m^2 (-10 to 1 dBsm), with multiple sources citing a central estimate of approximately 0.1–0.5 m^2 in optimal head-on aspect.

Table 2: Comparative RCS Values (X-band, Head-on, Clean Configuration)

Aircraft	**RCS (m^2)**	**RCS (dBsm)**
Su-57 (Felon)	0.1 – 0.5	-10 to -3
F/A-18 Super Hornet	0.8 – 1.0	-1 to 0
F-35 Lightning II	0.001 – 0.1	-30 to -10
F-22 Raptor	< 0.0001	$<$ -40

Note: RCS values are aspect, frequency, and configuration dependent. Values presented represent head-on X-band estimates for clean configurations with internal bays closed. Side and rear aspect RCS values are significantly higher across all platforms. At lower frequency bands (L, VHF), stealth effectiveness is reduced for all aircraft listed.

8.1.2 RCS Analysis

The Su-57's RCS reduction strategy differs fundamentally from Western stealth designs. Unlike the F-22 and F-35 which employ serpentine intake

ducts to block engine face radar returns, the Su-57 uses radar blockers and inlet guide vanes with RAM coating. While effective, this approach yields a higher frontal RCS than first-generation Western stealth designs.

At L-band frequencies, the Su-57's N036L wing leading edge arrays exploit this limitation in reverse — using the same frequency range where stealth is less effective to detect low-observable adversaries such as the F-22 and F-35.

Source: Nepryaev, A. et al. (2020). "The Calculated Data of the Radar Cross-Section Parameters of the 3D Model of the Sukhoi Su-57 Jet Fighter." ResearchGate. Available at: https://www.researchgate.net/publication/345140504

9 Armament

The Su-57 features a comprehensive internal weapons architecture preserving stealth characteristics across all mission profiles. Internal carriage eliminates external drag penalties, supporting supercruise capability and reducing radar signature during ingress.

9.1 Weapons Bay Architecture

The aircraft incorporates four internal weapons bays:

- **Two Main Tandem Bays** — Primary bays arranged in tandem between the engines, each with dimensions of approximately 4.4m × 0.9m, housing up to four beyond-visual-range missiles total.
- **Two Side Quick-Launch Bays** — Located at the wing roots, each housing one short-range air-to-air missile on a rapid-deploy launcher mechanism, publicly demonstrated at Dubai Airshow 2025.

Total internal hardpoints: 6. External hardpoints: 6 (stealth profile compromised when used).

9.2 Internal Cannon

- **9A1-4071K GSh-30-1** — 30mm autocannon embedded in the right LEVCON root, 150 rounds, maximum theoretical rate of fire 1,500 rounds/min, weighing approximately 50kg.

9.3 Air-to-Air Missiles

Table 3: Su-57 Air-to-Air Missile Loadout

Missile	Guidance	Range	Role
R-77M	Active radar	Medium-BVR	Primary BVR, main bays
R-74M2 (K-74M2)	Infrared	Short-range	WVR, side bays, 75° off-boresight
R-37M	Active radar	300km+	Very long range BVR

9.4 Air-to-Ground and Anti-Radiation

Table 4: Su-57 Air-to-Ground Loadout

Weapon	Range	Role
Kh-69 (Kh-59MK2)	∼300km	Precision strike, subsonic cruise missile
Kh-38M	Medium	Tactical air-to-ground, folding wings for internal carriage
Kh-58UShK	>200km	Anti-radiation SEAD, Mach 3.6, folding fins
Kh-35U	Medium	Anti-ship
KAB-250L	–	Electro-optically guided bomb
KAB-500	–	Guided bomb

9.5 Defensive Systems

- **101KS-O DIRCM** — Directed infrared countermeasures system, blinds hostile IR-guided missiles using focused infrared beam. Believed to be the first operational fitment of DIRCM on a fighter aircraft.
- **101KS-V IRST** — Passive infrared search and track with laser rangefinder.
- **L402 Himalayas** — Active jamming suite housed in tail cone.
- Chaff and flare dispensers.

9.6 Standard Combat Loadout

For a standard air superiority mission with stealth preserved:

- 4 × R-77M (main tandem bays)
- 2 × R-74M2 (side quick-launch bays)
- 1 × 30mm cannon (150 rounds)

10 Strengths and Limitations

10.1 Strengths

- **Supermaneuverability** — The combination of LEVCON vortex controllers and 3D thrust vectoring enables post-stall maneuvering at angles of attack exceeding $60 - 70°$, supporting maneuvers such as the Pugachev's Cobra, Kulbit, and Herbst maneuver. This provides a significant advantage in within-visual-range combat.
- **Counter-Stealth Capability** — The N036L L-band wing leading edge arrays exploit resonance effects to detect low-observable aircraft including the F-22 and F-35, partially offsetting the Su-57's own stealth disadvantage in BVR engagements.
- **Comprehensive Weapons Suite** — Internal carriage of diverse munitions across all mission profiles (BVR, WVR, SEAD, precision strike, anti-ship) with stealth preserved. The R-37M provides a 300km+ BVR engagement capability unmatched by Western counterparts.
- **DIRCM System** — The 101KS-O directed infrared countermeasures system, believed to be the first operational fitment on a fighter aircraft, provides active defense against IR-guided missiles.
- **Supercruise** — Sustained supersonic cruise at Mach 1.3 without afterburner, reducing fuel consumption and infrared signature during ingress.
- **Cost Advantage** — Unit cost estimated at approximately $35–50M USD, significantly lower than the F-35 (~$80M) and F-22 (~$150M+), making it competitive in export markets.

10.2 Limitations

- **Stealth Inferiority** — The Su-57's RCS of 0.1–0.5 m^2 is significantly higher than the F-22 (<0.0001 m^2) and F-35 (˜0.001 m^2). The absence of serpentine intake ducts, panel gaps, and external sensor protrusions contribute to higher observability than Western fifth-generation designs.

- **Engine Limitations** — The AL-41F1 engine represents an evolution of the Su-35's powerplant rather than a clean-sheet fifth-generation design. The definitive Izdeliye 30 engine, promising 176 kN thrust and 20% improved fuel efficiency, remains in development as of 2026.

- **Production Rate** — As of early 2026, approximately 30 airframes have been delivered to the Russian Aerospace Forces, far below the stated target of 76 by 2027–2028. Only 2 new airframes were identified in 2025, reflecting persistent industrial capacity constraints and sanctions impacts on avionics supply chains.

- **Sensor Fusion Gap** — While the Sh121 radar suite is capable, reports suggest the Su-57's sensor fusion and data-link architecture lags behind the F-35's integrated battlespace picture, resulting in higher pilot workload in contested environments.

- **Limited Combat Record** — Su-57 deployments in Ukraine have been limited, with aircraft operating exclusively from standoff ranges well behind the front line. No large-scale operational deployment has been observed, limiting independent assessment of real-world combat effectiveness.

- **Export Failure** — India withdrew from the joint FGFA program in 2018 citing performance and technology transfer concerns. Algeria remains the only reported export customer, with delivery timelines uncertain given domestic production constraints.

10.3 Overall Assessment

The Su-57 represents a genuine fifth-generation capability, distinguished by world-class supermaneuverability, a unique counter-stealth detection architecture, and a comprehensive internal weapons suite. However, it reflects

a fundamentally different design philosophy from Western counterparts — prioritising agility, versatility, and cost over the deep low-observable integration that defines the F-22 and F-35. In a beyond-visual-range engagement against Western fifth-generation fighters, the Su-57's higher RCS places it at a detection disadvantage. In a within-visual-range engagement, its supermaneuverability and LEVCON-TVC combination likely provide a decisive advantage.

The program's primary constraint is not aerodynamic or avionics capability, but industrial — the inability to produce sufficient airframes to translate genuine technical achievement into strategic air power mass.

11 Conclusion

This paper has presented a comprehensive aerodynamic and systems analysis of the Sukhoi Su-57 (Felon) fifth-generation multirole fighter, employing NASA DATCOM methodology applied to publicly available geometric and performance data.

11.1 Aerodynamic Findings

The following performance values were derived from first principles:

- **Lift Curve Slope:** $C_{L_\alpha} = 2.728$ /rad (DATCOM method, half-chord sweep 38°)
- **Subsonic Drag Force:** $D \approx 60.8$ kN at sea level, $V = 206$ m/s, $C_D = 0.0297$
- **Instantaneous Turn Rate:** $\dot{\psi} \approx 24.4°$/s at 206 m/s, limited by pilot G-tolerance of 9g rather than aerodynamic or structural limits. LEVCON and TVC augmentation extend control authority at this load factor but do not meaningfully increase g beyond physiological limits.
- **Sustained Turn Rate:** $\dot{\psi} \approx 17.8°$/s at 154.3 m/s (300 kts) with full LEVCON and TVC augmentation at published thrust figures.
- **Rate of Climb:** $ROC \approx 176.7$ m/s (34,780 ft/min) at TWR = 1.09, sea level, $V = 206$ m/s. Maximum emergency power yields $ROC \approx 183.6$ m/s.

11.2 Performance Discrepancy Observation

A consistent pattern was identified between derived performance values and figures widely cited in open literature:

Table 5: Derived vs. Cited Performance Figures

Parameter	**Derived**	**Cited**	**Discrepancy**
Sustained Turn Rate	17.8°/s	$25 - 30°$/s	∼40%
Rate of Climb	176.7 m/s	350–384 m/s	∼100%

Critically, published component specifications — thrust (142.2 kN per engine) and thrust-to-weight ratio (1.09) — are internally consistent with each other. The discrepancies emerge specifically in integrated system-level performance figures. This suggests that widely circulated ROC and turn rate figures in open literature may originate from unverified secondary sources rather than official Sukhoi documentation, or may reflect performance achievable only under conditions not fully specified (altitude, weight, configuration).

Achieving the cited sustained turn rate of $25 - 30°$/s at 300 kts would require approximately 427 kN total thrust — 50% above published AL-41F1 figures — which is inconsistent with the published TWR of 1.09. The cited ROC of 350–384 m/s would require approximately 506–549 kN total thrust, approximately double published figures, which is physically implausible with any known engine variant fitted to the Su-57.

11.3 Overall Assessment

The Su-57 represents a genuine and significant aerodynamic achievement. Its LEVCON-TVC supermaneuverability architecture is unique among operational fifth-generation fighters, enabling post-stall control authority at angles of attack exceeding $60 - 70°$. Its N036L counter-stealth L-band arrays represent an innovative asymmetric response to Western low-observable designs. Its comprehensive internal weapons suite supports the full spectrum of air superiority, SEAD, and precision strike missions while preserving stealth.

The aircraft's primary operational constraint is not technical capability but industrial — production rates remain insufficient to translate genuine aerodynamic and systems excellence into meaningful strategic air power mass.

The derived performance figures presented in this paper, while lower than commonly cited values, still represent a highly capable platform competitive with any fourth-generation fighter and formidable in within-visual-range engagements against fifth-generation adversaries. The Su-57 should be evaluated on the basis of verified, internally consistent data rather than figures of uncertain provenance.

All values in this paper are derived from publicly available data using NASA DATCOM methodology and standard aerodynamic equations. They represent engineering approximations and should not be treated as definitive specifications. See supplementary calculations document for full derivation details.

11.4 Known Limitations and Future Work

The following limitations are acknowledged in this version and will be addressed in subsequent revisions:

- **LEVCON Geometric Data** — LEVCON contribution to lift and sustained turn rate augmentation ($\delta n_{LEVCON} = +0.627g$) was estimated assuming a +15% C_L increment. Precise derivation requires detailed LEVCON planform geometry, which is not available in the public domain. This estimate will be refined in v2 if geometric data becomes available through the DCS development community.

- **Izdeliye 30 Engine Not Modelled** — All propulsion calculations in this paper use AL-41F1 (117S) specifications. The Izdeliye 30 engine, promising approximately 176 kN thrust per engine and 20% improved fuel efficiency, remains in development as of 2026 and lacks verified public performance data. A dedicated performance comparison will be included in v2 when data becomes available.

- $C_{L\alpha}$ **Open Literature Benchmark Absent** — Table 1 does not include an open literature value for $C_{L\alpha}$, as no independently reported figure was identified for the Su-57 specifically. A comparable delta-wing DATCOM estimate will be sourced for v2.

- **Wave Drag Model Simplified** — $C_{D_{wave}}$ at supercruise was treated as a constant (0.02) rather than derived from a Mach-dependent wave drag model (e.g. Lock's or Kármán Ogive approximation). A full transonic drag rise model will be incorporated in v2.

- **Weapon Bay Induced Drag Assumption** — The bay-open drag calculation in Section 7.3.1 uses the clean-configuration induced drag coefficient C_{Di}. In practice, C_{Di} varies with flight condition during weapon release. This will be refined in v2 with a condition-specific analysis.

- **Control Surface Derivatives from Open Literature** — $C_{l_{\delta_a}}$, $C_{n_{\delta_r}}$, C_{l_p}, and C_{n_r} are retained from open literature rather than independently derived, as first-principles derivation requires detailed control surface geometry unavailable in the public domain.

References

[1] Blake, W.B. (1998). *Missile DATCOM: User's Manual.* NASA Technical Report 19950005762. NASA Technical Reports Server. Available at: https://ntrs.nasa.gov/citations/19950005762

[2] Various. *Sukhoi Su-57 Technical Specifications.* Public domain sources including Jane's All the World's Aircraft, Britannica, and open aerospace literature.

[3] Nepryaev, A. et al. (2020). *The Calculated Data of the Radar Cross-Section Parameters of the 3D Model of the Sukhoi Su-57 Jet Fighter.* ResearchGate. Available at: https://www.researchgate.net/publication/345140504

[4] The Aviationist. (2025). *Russia Shows Off Su-57's Internal Weapons Bays for the First Time.* Available at: https://theaviationist.com/2025/11/11/russia-shows-off-su-57s-internal-weapons-bays/

[5] Encyclopaedia Britannica. (2025). *Sukhoi Su-57.* Available at: https://www.britannica.com/technology/Sukhoi-Su-57

[6] Anderson, J.D. (2001). *Fundamentals of Aerodynamics.* 4th ed. McGraw-Hill.

[7] Raymer, D.P. (2018). *Aircraft Design: A Conceptual Approach.* 6th ed. AIAA Education Series.

[8] Etkin, B. and Reid, L.D. (1996). *Dynamics of Flight: Stability and Control.* 3rd ed. Wiley. [Stability derivatives, Cm_alpha, Cm_q, lateral-directional coefficients]

[9] McCormick, B.W. (1995). *Aerodynamics, Aeronautics and Flight Mechanics.* 2nd ed. Wiley. [Turn rate analysis, ROC methodology, performance equations]

[10] Hull, D.G. (2007). *Fundamentals of Airplane Flight Mechanics.* Springer. [Sustained and instantaneous turn rate derivations]